Seminar report (WP4 proving the proposed concept)

The seminar was held within Work Package WP4 (Proving the proposed concept). Its subject was the presentation of the implemented concept of statistical monitoring of quantisation in neural networks, together with its experimental validation. The aim of the meeting was to summarise the current state of the work — from the theoretical foundation through the core source code and the proposed training strategy to the experimental results — and to define the next steps leading to the project's final report. This report summarises the technical subject matter that was presented.

1. Background and problem

Quantisation of neural networks reduces the number of bits used to represent the model's weights. This yields substantial savings in memory and computation, which is essential for deploying models on resource-constrained devices, but it can also degrade the model's accuracy. A fundamental problem is that standard metrics such as accuracy and the loss function reveal only the final impact of compression, not its cause, and do not make it possible to observe what is happening to the model at the level of its weight distribution. The central question of the project is therefore whether the impact of quantisation on the weight distribution can be tracked statistically and continuously — already during training — thereby providing a cheap and interpretable indicator of degradation.

2. The statistical similarity index

The core of the proposed concept is a statistical similarity index, which quantifies the difference between the weight distribution of the reference (full-precision, FP32) model and that of the quantised model. Three statistical measures are used to quantify this difference: the Kullback–Leibler divergence, which is sensitive and unbounded and reveals information loss well; the Jensen–Shannon divergence, which is symmetric and bounded to the interval [0, 1], and is therefore suited to automated decision-making and thresholding (it is the default in the implementation); and cosine similarity, which provides a rough comparison of the overall orientation of the distributions. The index can be computed globally (over the weights of the whole model) or per layer, which makes it possible to identify the layer most sensitive to quantisation. The index thus acts as a “thermometer” of weight-distribution shape — the higher it is, the more the quantised weights have deviated from the originals.

3. Implementation — core of the source code

The core of the source code was presented, comprising four interconnected parts. The first is the characterisation of weights through distributional moments, Shannon and differential entropy, and histograms with consistent bin edges, which enables comparable measurements. The second is post-training quantisation, which snaps the weights onto a grid of 2*b* levels. The third is genuine quantisation-aware training (QAT), in which the forward pass uses quantised weights and the gradient is propagated through a straight-through estimator; this is realised by the custom QuantConv2D and QuantDense layers. The fourth — and the most important for robustness — is per-channel quantisation, which assigns a separate scale to each output channel instead of a single scale for the whole kernel. This approach is essential for deep networks, where naive per-tensor quantisation fails.

4. The new training strategy

The output of the work is a new training strategy that integrates the statistical index directly into the genuine-QAT process and uses it to steer training. The strategy rests on three components. The monitoring component computes the value of the index after every epoch and displays it live, so that the evolution of the quantisation state is continuously visible. The early-stopping component halts training at the point where both the index and accuracy stabilise, thereby saving compute. The automatic bit-width selection component chooses the lowest bit-width at which the index remains below a set threshold. These components are combined into a single training procedure.

5. Experimental validation of the concept

The focus of the seminar was the experimental validation, which constitutes the substance of Work Package WP4. The bit-width sensitivity analysis showed that the index tracks quantisation severity almost perfectly and monotonically and predicts the drop in accuracy, consistently across three datasets of varying difficulty. Tracking the dynamics during QAT demonstrated that throughout training the index holds steady at the level corresponding to the given bit-width — which proves that the weights remain genuinely quantised and that training does not "drift" them back to full precision — while accuracy itself is maintained, or recovered, relative to naive post-training quantisation.

Generalisation to large models (ImageNet) also produced an important and honestly presented finding about the limits of the method. While the MobileNetV2 architecture confirmed the expected pattern, the ResNet50 architecture revealed the fragility of naive per-tensor quantisation for deep networks: the accuracy drop occurred already at a higher bit-width, yet the global weight index did not capture it, because the degradation did not originate in the shape of the weight distribution. This finding directly motivated the move to per-channel quantisation and points to the need for complementary monitoring at the level of layers and activations.

6. Findings and limitations

The presented results indicate that the statistical index works in two modes — before training as a predictor of the severity of quantisation, and during training as a stable monitor of it — and is at the same time independent of the particular dataset and of the choice of metric. The limitations were also stated honestly: the experiments rely on smaller models and a limited number of initialisations, per-channel quantisation is necessary for deep networks, and the global index cannot capture degradation that is driven by activations and is not reflected in the weight distribution. These limitations define the direction of further research.

7. Next steps

Following the seminar, the work will continue with the completion of the outputs of Work Packages WP3 and WP4 and with the preparation of the project's final report (D17). The directions identified for future work are monitoring at the level of individual layers and activations, a principled mixed-precision design, validation on larger models with a higher number of initialisations, and a real integer (int8 / TFLite) export of models.

Conclusion

The seminar fulfilled its aim — it presented the implemented concept of statistical monitoring of quantisation and substantiated it with experimental validation, including an honest statement of the limits of the method. The meeting thus established a coherent and well-supported basis for the final phase of the project and for the preparation of the final report.

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